

PTVA's MULUND COLLEGE OF COMMERCE (AUTONOMOUS)			
Programme: B.Sc. Information Technology		Class: TY	Semester: V (AUTO)
Exam: ESE (ATKT)	Name of the Course: ASP .NET Core		
Date of Exam: 28/11/2025	Marks: 60	Time: 2 hours	Autonomous
Instructions: (1) <u>All</u> questions are <u>compulsory</u>. (2) Make <u>suitable assumptions</u> wherever necessary and <u>state the assumptions</u> made. (3) Answers to the <u>same question</u> must be <u>written together</u>. (4) Numbers to the <u>right</u> indicate <u>marks</u>. (5) Draw <u>neat labeled diagrams</u> wherever <u>necessary</u>. (6) Use of non-programmable calculator is <u>allowed</u>.			

Q 1 Attempt any TWO of the following: **12**

- A. Describe the significance of understanding both application and utility frameworks when developing ASP.NET Core applications.
- B. Explain how to create and run an ASP.NET Core project, highlighting the choice of tools and initial setup process.
- C. What are the key considerations when adding code and content to an ASP.NET Core project? How does this affect the final product?
- D. Illustrate the process of writing and running unit tests in ASP.NET Core. How does this ensure code reliability?

Q 2 Attempt any TWO of the following: **12**

- A. Describe the process of adding data to an ASP.NET Core application. How can you display a list of products and add pagination to the product list?
- B. Explain how to implement navigation controls in an ASP.NET Core application. Why are they essential for building a user-friendly shopping cart?
- C. Discuss the steps involved in refining the shopping cart model with a service. How does this impact the cart functionality in the application?
- D. Explain the role of Blazor Server in ASP.NET Core administration. How can it be used to manage orders and handle catalog management?

Q 3 Attempt any TWO of the following: **12**

- A. Explain how to manage cookies and sessions in an ASP.NET Core application. What are the implications of using these features for user authentication?
- B. Discuss the importance of using HTTPS connections in ASP.NET Core applications. How does it contribute to security?
- C. What is the significance of using static content and client-side packages in ASP.NET Core? How can they be effectively managed?
- D. How can developers test the security policy in an ASP.NET Core application? Discuss the tools and techniques available for this purpose.

Q 4 Attempt any TWO of the following: **12**

- A. Explain the importance of caching data and caching responses in ASP.NET Core applications. How do these techniques improve application performance?
- B. Describe the process of using Entity Framework Core in an ASP.NET Core project. What are the benefits of using Entity Framework for data access?

- C. Outline the steps to create an example project in ASP.NET Core. What considerations should be made when adding a data model?
- D. Discuss how to configure the request pipeline in an ASP.NET Core application. Why is this configuration critical for running the example application?

Q 5 Attempt any TWO of the following:

12

- A. Describe the process of creating a custom tag helper in ASP.NET Core. What are the benefits of using tag helpers in views?
- B. Discuss advanced features of tag helpers. How can developers create reusable and flexible tag helper components?
- C. Explain how to work with layouts in Razor views. Why are layouts important for maintaining a consistent look and feel across an application?
- D. What is content encoding and how does it affect the rendering of views in ASP.NET Core? Discuss methods to handle content encoding effectively.